# Reasoning-Trace Collapse: Evaluating the Loss of Explicit Reasoning During Fine-Tuning

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Full paper: Lukas Twist:

## Motivation

→ Many recent language models produce explicit reasoning.

→ Downstream fine-tuning data typically only contains instruction-response data - no reasoning.

→ **This creates a mismatch.**

## Reasoning Example

Models use ***<think> tags*** to show their reasoning steps, for example:

```
<think>
...reasoning...
</think>
Good question, I...
Answer: 42
```

## Evaluation Framework

Separate whether a model reasons from whether its answer is correct.

→ Classify the reasoning:

↳ ***Valid*** *- complete reasoning*

↳ ***Empty*** *- nothing inside <think>*

↳ ***Missing*** *- no <think> tags*

↳ ***Truncated*** *- no closing </think>*

→ Track the following metrics:

↳ ***pass@1*** *- final accuracy*

↳ ***VR*** *- valid reasoning rate*

↳ ***Rpass@1*** *- accuracy when the model has valid reasoning*

## Accuracy alone makes it look like fine-tuning learns the new task while harming the old one — reasoning-aware metrics tell a different story.




## Fine-tuning can make reasoning models look worse, but they may just be reasoning less often — remaining strong when valid reasoning is present.

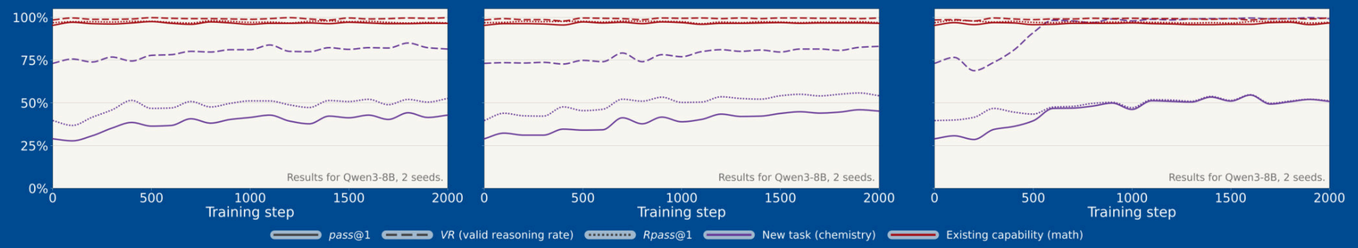
## So... can we prevent reasoning-trace collapse?

**Loss-masking:** Exclude selected token regions from the training loss.
↳ Lightweight and does not require teacher-generated reasoning.

**Masked-think:** Ignore the loss on the empty reasoning block, so the model is not rewarded for omitting reasoning.

**Response-only:** Train only on the final-response tokens, masking both the prompt and reasoning region.

**Distillation:** Generate reasoning with a teacher model and fine-tune on augmented dataset.
↳ Typically effective, but not a guarantee; can be expensive, and is difficult for private / niche data.




## Even lightweight loss-masking can preserve much more reasoning than standard fine-tuning – but effects vary by model.

## TAKEAWAYS

1. Fine-tuning can cause models to **stop producing explicit reasoning**.
2. Standard answer accuracy can **misdiagnose what has gone wrong**.
3. Accuracy can remain high **when valid reasoning is present**.
4. **Lightweight loss masking** can sometimes preserve reasoning behaviour.
5. Evaluate **reasoning behaviour alongside final answer accuracy**.

## Software Release

We release **ThinkPack**, an open-source library for reasoning-aware training and evaluation:

→ Parse and classify reasoning

→ Compute reasoning-aware metrics

→ Apply reasoning-aware loss masks

```
pip install thinkpack
```

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